

Income Tax Evaluation & Maintenance System

Project Documentation & User Guide

Version: 2.0 | **Last Updated:** July 2026

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1. Project Overview

The **Income Tax Evaluation & Maintenance System** is a comprehensive desktop and web-based application designed to streamline the management of income tax records, client information, and financial accounts for tax consultants, Chartered Accountants (CAs), and businesses.

The system automates the calculation of income tax, maintains detailed client records, manages trading accounts, profit & loss statements, and balance sheets — all integrated into a single, user-friendly platform.

Key Highlights:

- Complete client lifecycle management
- Automated tax computation based on Indian tax rules
- Integrated financial accounting (Trading, P&L, Balance Sheet)
- Secure role-based access control
- Both Desktop (VB.NET + Oracle) and Web (HTML/JS) versions available
- Local browser storage for web demo — no server required

2. Purpose & Benefits

Why This Project?

Traditional tax management involves scattered Excel sheets, manual calculations, and disconnected records. This system brings everything together:

- **Eliminates Manual Errors:** Automated tax calculations reduce human error
- **Saves Time:** Quick client lookup, auto-filled forms, and instant reports
- **Audit Trail:** Track who created/updated records and when
- **Centralized Data:** All client and tax information in one secure database
- **Compliance Ready:** Structured data aligned with Indian tax filing requirements
- **Multi-User Access:** Role-based login for teams

3. Technology Stack

Desktop Application

Component	Technology
Language	VB.NET (.NET Framework 4.8)
Database	Oracle Database (Oracle.ManagedDataAccess)
UI Framework	Windows Forms (WinForms)
Architecture	3-Tier (Presentation, Business Logic, Data Access)
IDE	Visual Studio 2022

Web Demo Application

Component	Technology
Frontend	HTML5, CSS3, JavaScript (Vanilla)
Storage	Browser LocalStorage (no backend required)
Hosting	GitHub Pages
URL	https://ankitgupta1729.github.io/IncomeTaxEvaluationSystem-WebDemo/

Architecture Pattern: The desktop app follows a 3-tier architecture:

1. **Presentation Layer:** Windows Forms (UI)
2. **Business Logic Layer:** Model classes & Validation module
3. **Data Access Layer:** OracleDataAccess class with parameterized queries

4. System Architecture

The system is organized into the following modules:

- **Data Layer:** OracleDataAccess.vb — Handles all database operations

- **Models:** ClientRecord, IncomeTaxRecord, TradingAccount, PLAccount, BalanceSheet
- **Forms:** LoginForm, MainForm, ClientRecordForm, IncomeTaxRecordForm, TradingAccountForm, PLAccountForm, BalanceSheetForm, ReportsForm
- **Modules:** Validation.vb — Input validation and business rules
- **Reports:** ReportGenerator.vb — CSV export functionality

5. Database Schema

The Oracle database consists of the following core tables:

Client Record (CLIENT_RECORD)

Column	Type	Description
CLIENT_CODE	VARCHAR2	Unique client identifier
FULL_NAME	VARCHAR2	Client's full name
PAN	VARCHAR2	Permanent Account Number
GENDER	VARCHAR2	Male/Female/Other
CITY, STATE, PINCODE	VARCHAR2	Address details
MOBILE, EMAIL	VARCHAR2	Contact information
ASSESSMENT_YEAR	VARCHAR2	Current assessment year

Income Tax Record (INCOME_TAX_RECORD)

Column	Description
CLIENT_CODE	Foreign key to client
ASSESSMENT_YEAR	Tax assessment year
INCOME_FIELDS	Salary, House Property, Business, Capital Gains, Other Sources
DEDUCTIONS	80C, 80D, 80G, Other Deductions, Total Deductions
TAX_CALCULATION	Taxable Income, Tax on Income, Rebate, Surcharge, Cess, Total Tax
PAYMENTS	TDS, Advance Tax, Self Assessment Tax, Refund

Other Tables

- **TRADING_ACCOUNT:** Opening balance, purchases, sales, closing stock, expenses
- **PL_ACCOUNT:** Trading account, purchases, gross profit, expenses, net profit
- **BALANCE_SHEET:** Liabilities (share capital, loans, reserves) and Assets (fixed assets, investments, debtors)
- **APP_USERS:** User authentication and role management

6. Features & Modules

6.1 Client Management

- Add, edit, delete client records
- Search clients by code, name, or PAN
- Validate PAN format (ABCDE1234F), mobile, email, pincode
- Store complete client profile with address and contact details
- Track created/updated by and timestamps

6.2 Income Tax Computation

- Record income from multiple sources: Salary, House Property, Business, Capital Gains, Other Sources
- Apply deductions under Chapter VI-A: 80C, 80D, 80G, Other Deductions
- Auto-calculate: Total Income, Taxable Income, Tax on Income
- Apply rebate u/s 87A, surcharge, and health/education cess (4%)
- Track TDS, Advance Tax, Self Assessment Tax
- Calculate refund due or balance tax payable
- Support for filing status: Pending, Filed, Verified, Rejected

6.3 Trading Account

- Record opening balance, purchases, purchase returns, sales, sales returns
- Auto-calculate: Gross Purchases, Closing Stock, Direct Expenses, Gross Profit, Net Profit
- Track creditors and debtors

6.4 Profit & Loss Account

- Opening Stock, Purchases, Closing Stock
- Auto-calculate COGS: $\text{Opening Stock} + \text{Purchases} - \text{Closing Stock}$
- $\text{Gross Profit} = \text{COGS}$
- Track all expenses: Administrative, Selling, Employee Benefits, Finance Costs, Depreciation, etc.
- Auto-calculate Total Expenses and Net Profit
- Other Income: Interest, Dividends

6.5 Balance Sheet

- Liabilities: Share Capital, Reserves & Surplus, Secured/Unsecured Loans, Current Liabilities
- Auto-calculate Total Liabilities
- Assets: Fixed Assets, Investments, Inventory, Debtors, Cash & Bank, Other Current Assets
- Auto-calculate Total Assets
- Contingent Liabilities tracking

6.6 Reports & Export

- Generate reports for Clients, Tax Records, Trading, P&L, Balance Sheet
- Filter reports by specific client
- Export data to CSV format for Excel analysis

6.7 Security & Audit

- Role-based login system
- User authentication against APP_USERS table
- Audit trail for all create/update/delete operations
- Session management

7. Getting Started (Desktop Application)

Prerequisites

Requirement	Details
Operating System	Windows 10/11
Framework	.NET Framework 4.8
IDE	Visual Studio 2022 (Community/Professional/Enterprise)
Database	Oracle Database 11g/12c/19c/21c
Oracle Client	Oracle.ManagedDataAccess NuGet package (v4.122.1)

Installation Steps

Step 1: Clone or Download the Repository

```
git clone https://github.com/ankitgupta1729/AI-ML-DS.git
cd AI-ML-DS/18.Utilities/Project/IncomeTaxEvaluationSystem
```

Step 2: Set Up Oracle Database

```
-- Connect to Oracle as SYSDBA and run:
@Database/Schema.sql
```

This creates all required tables: CLIENT_RECORD, INCOME_TAX_RECORD, TRADING_ACCOUNT, PL_ACCOUNT, BALANCE_SHEET, APP_USERS.

Step 3: Update Connection String

Open `App.config` and update the connection string:

```
<connectionStrings>
  <add name="IncomeTaxDB"
        connectionString="User Id=YOUR_USER;Password=YOUR_PASS;Data Source=localhost:1521/XEPDB1"
        providerName="Oracle.ManagedDataAccess.Client" />
```

```
</connectionStrings>
```

Step 4: Restore NuGet Packages

In Visual Studio: Right-click solution → Restore NuGet Packages

Or via Package Manager Console:

```
Install-Package Oracle.ManagedDataAccess -Version 4.122.1
```

Step 5: Build and Run

Set `LoginForm` as startup form and press F5.

8. Web Demo Guide

Access URL

<https://ankitgupta1729.github.io/IncomeTaxEvaluationSystem-WebDemo/>

Demo Credentials

Field	Value
User ID	admin
Password	admin123

Features

- No installation required — runs entirely in browser
- Data stored locally in browser's LocalStorage
- Fully functional client and tax record management
- Dark mode support
- Responsive design for mobile and desktop
- Keyboard shortcuts (Enter to login)

Resetting Demo Data

Click **"Reset Demo Data"** in the Quick Actions panel on the Dashboard to clear all locally stored data and start fresh.

9. Navigation & Usage

Login Screen

Enter credentials to access the system. The login screen is clean and distraction-free.

Dashboard

After login, the Dashboard provides:

- Quick stats: Total Clients, Tax Records, Pending Filings, Total Tax Liability
- Recent Activity log

- Quick Action buttons for common tasks

Sidebar Navigation

The left sidebar provides quick access to all modules:

-  **Dashboard** — Home/Overview
-  **Clients** — Client Record Management
-  **Income Tax** — Tax Record Entry & Computation
-  **Trading** — Trading Account Management
-  **P&L Account** — Profit & Loss Statements
-  **Balance Sheet** — Assets & Liabilities
-  **Reports** — Generate & Export Reports

Client Records Page

Manage client information with full CRUD operations:

1. Click **+ New Client** to open the form
2. Fill in Client Code, Name, PAN (format: ABCDE1234F), Gender, City, Mobile, Assessment Year
3. Click **Save** to store the record
4. Use the search box to filter clients
5. Click **Edit** or **Del** buttons to modify or remove records

Income Tax Records Page

1. Select a client from the dropdown
2. Enter Assessment Year
3. Fill in income sources: Salary, House Property, Business, Capital Gains, Other Sources
4. Enter deductions: 80C, 80D, 80G, Other Deductions
5. System auto-calculates: Total Income, Deductions, Taxable Income, Tax on Income, Rebate, Cess, Total Tax Liability
6. Enter payment details: TDS, Advance Tax, Self Assessment Tax
7. System calculates Refund Due or Balance Tax
8. Set Filing Status and Return Type
9. Click **Save**

Trading Account Page

1. Select client and enter Assessment Year
2. Enter Opening Balance, Purchases, Purchase Returns, Sales, Sales Returns
3. System auto-calculates Gross Purchases, Closing Stock, Direct Expenses, Gross Profit, Other Income, Net Profit
4. Enter Creditors and Debtors
5. Click **Save Trading**

P&L Account Page

1. Select client and enter Assessment Year
2. Enter Opening Stock, Purchases, Closing Stock
3. System calculates COGS and Gross Profit
4. Enter all expense categories
5. System calculates Total Expenses and Net Profit
6. Enter Other Income items
7. Click **Save P&L**

Balance Sheet Page

1. Select client and enter Assessment Year
2. Enter Liabilities: Share Capital, Reserves & Surplus, Secured/Unsecured Loans, Current Liabilities

3. System auto-calculates Total Liabilities
4. Enter Assets: Fixed Assets, Investments, Inventory, Debtors, Cash & Bank, Other Current Assets
5. System auto-calculates Total Assets
6. Enter Contingent Liabilities if any
7. Click **Save Balance Sheet**

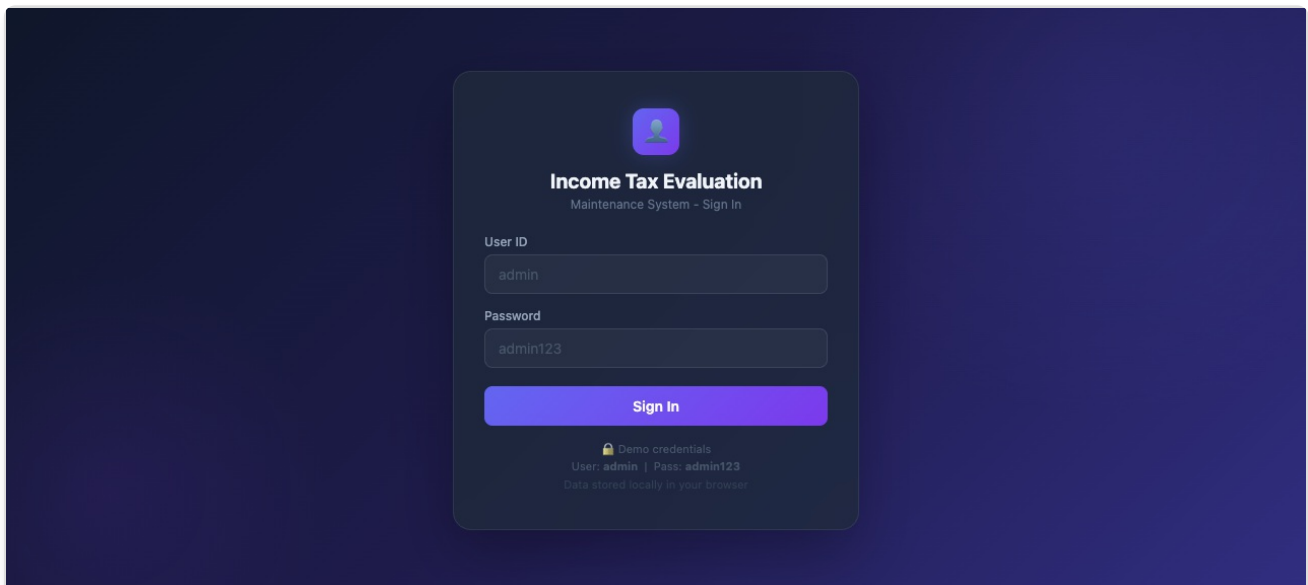
Reports Page

1. Select report type: Clients, Income Tax Records, Trading Accounts, P&L Accounts, or Balance Sheets
2. Optionally select a specific client to filter
3. Click **Generate** to view the report table
4. Click **CSV** to export the report to Excel-compatible format

10. Screenshots

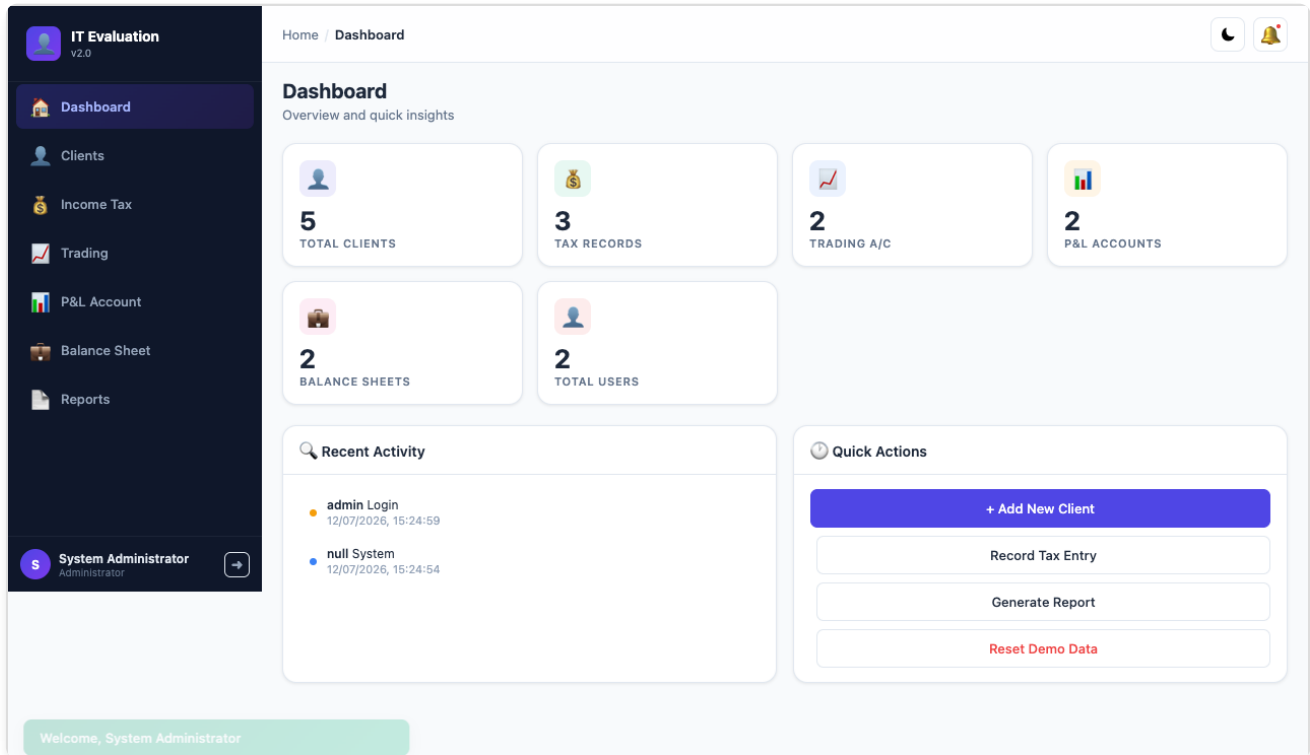
10.1 Login Screen

Clean, secure login interface with demo credentials hint.



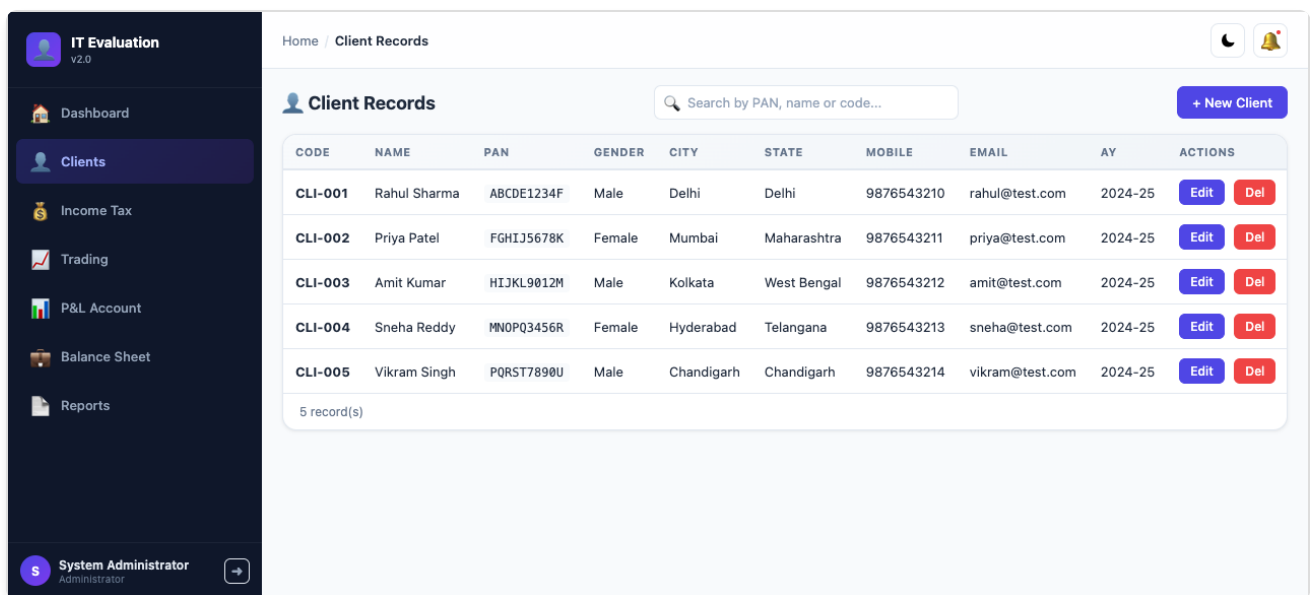
10.2 Dashboard

Overview with quick stats, recent activity, and quick action buttons.



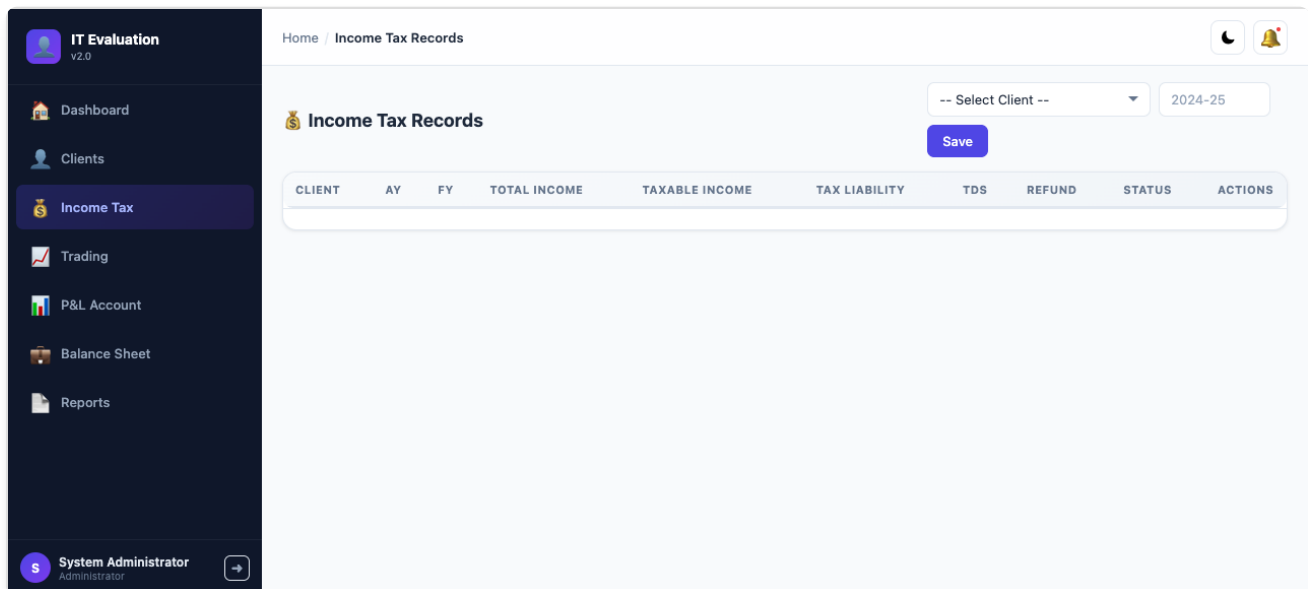
10.3 Client Records

Full client management with search, add, edit, and delete functionality.



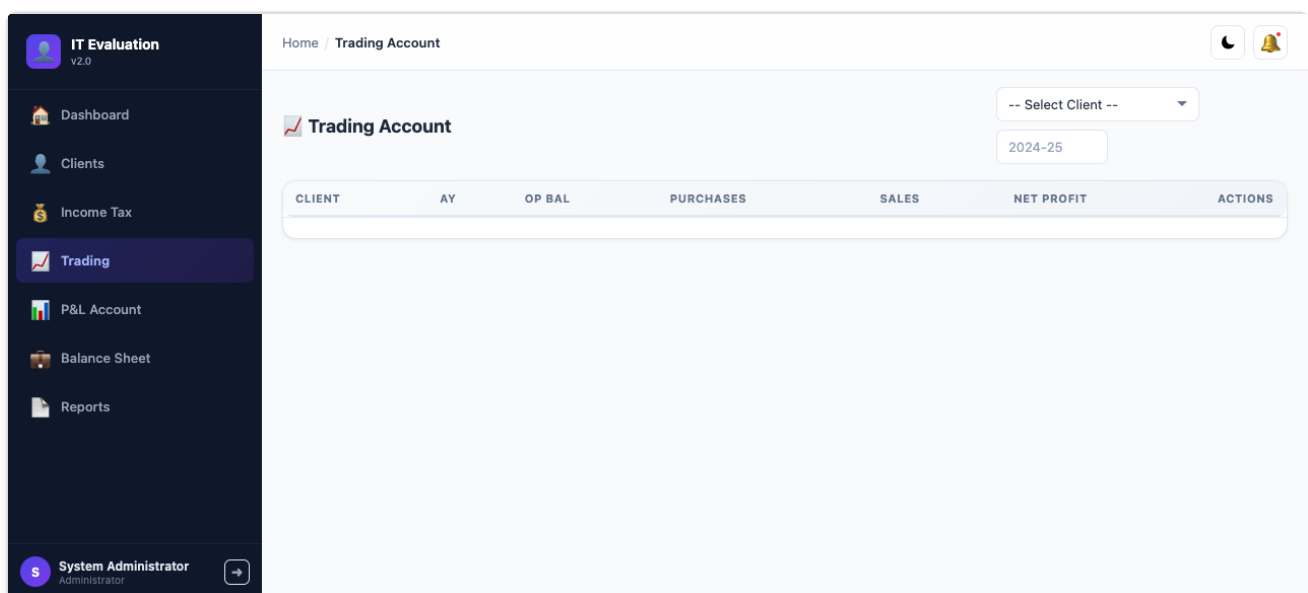
10.4 Income Tax Records

Comprehensive tax computation with income sources, deductions, and auto-calculated tax liability.



10.5 Trading Account

Track opening balance, purchases, sales, and auto-calculate profit/loss.



10.6 P&L Account

Profit & Loss statement with COGS, expenses, and net profit calculation.

IT Evaluation
v2.0

Dashboard

Clients

Income Tax

Trading

P&L Account

Balance Sheet

Reports

S

System Administrator

Administrator

Home / P&L Account

2024-25

-- Select Client --

Save P&L

CLIENT	AY	COGS	GROSS PROFIT	EXPENSES	NET PROFIT	ACTIONS
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10.7 Balance Sheet

Assets and liabilities management with auto-totals.

IT Evaluation
v2.0

Dashboard

Clients

Income Tax

Trading

P&L Account

Balance Sheet

Reports

S

System Administrator

Administrator

Home / Balance Sheet

2024-25

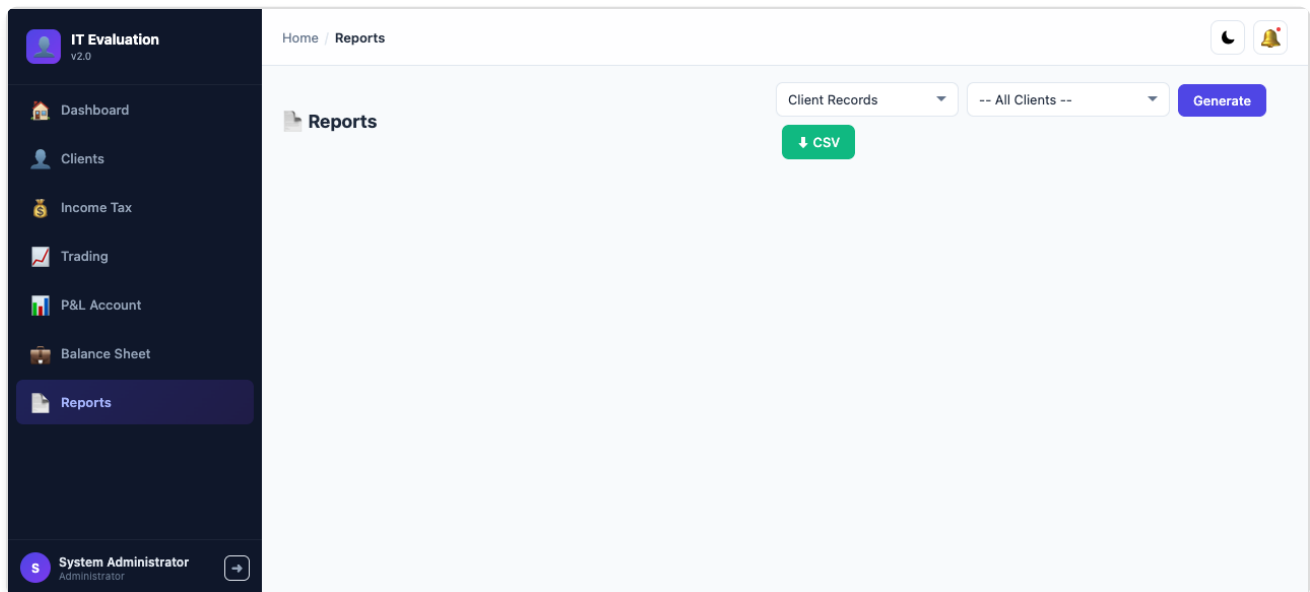
-- Select Client --

Balance Sheet

CLIENT	AY	LIABILITIES	ASSETS	CONTINGENT	ACTIONS
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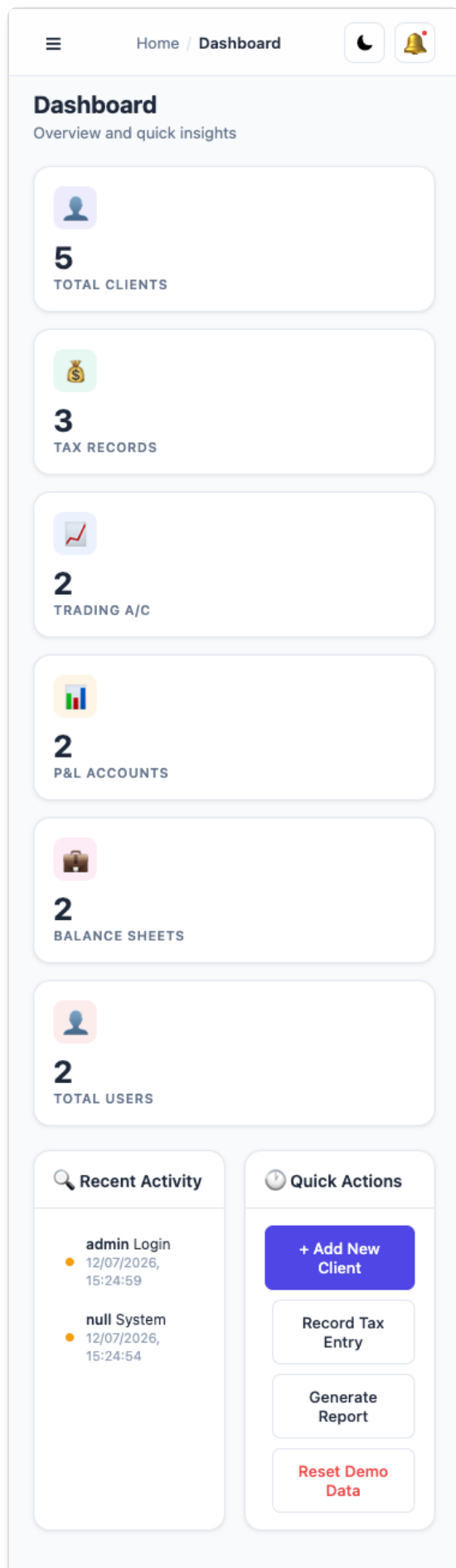
10.8 Reports

Generate and export various reports in CSV format.

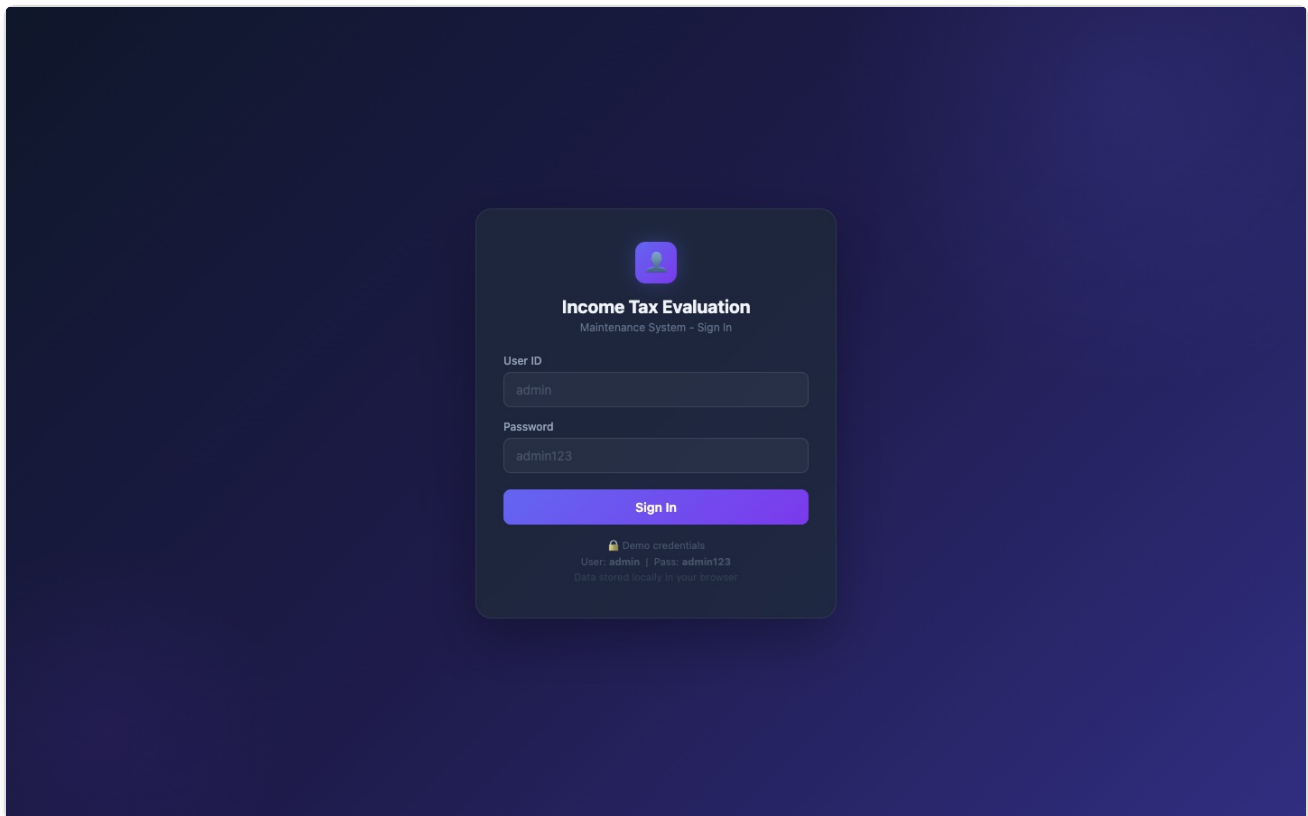


10.9 Mobile View

Fully responsive design for mobile devices.



Dark mode support for comfortable viewing in low light.



11. Troubleshooting

Desktop App Issues

Issue	Solution
Oracle connection error	Verify connection string in App.config, ensure Oracle DB is running
NuGet package restore fails	Run <code>Update-Package -reinstall</code> in Package Manager Console
Build errors about missing designer	Ensure all .Designer.vb files are present and included in project
Login fails with valid credentials	Check APP_USERS table has correct data and IS_ACTIVE='Y'

Web Demo Issues

Issue	Solution
Login not working	Use exactly <code>admin / admin123</code> (case-sensitive)
Data not persisting	Check browser LocalStorage is enabled (not incognito/private mode)
Page looks broken	Clear browser cache and reload (Ctrl+Shift+R)
Calculations showing NaN	Ensure all numeric fields have valid numbers

12. Future Enhancements

- PDF report generation with charts and graphs
 - Email integration for sending tax computation summaries
 - Bulk import/export of client data via Excel
 - Multi-year comparison reports
 - Integration with GST and TDS filing portals
 - Cloud-based deployment with real-time sync
 - Mobile native app (Android/iOS)
 - AI-powered tax optimization suggestions
 - Multi-language support
-

Contact & Support

For issues, feature requests, or contributions, please reach out to the development team.

Repository: <https://github.com/ankitgupta1729/AI-ML-DS>

Web Demo: <https://ankitgupta1729.github.io/IncomeTaxEvaluationSystem-WebDemo/>